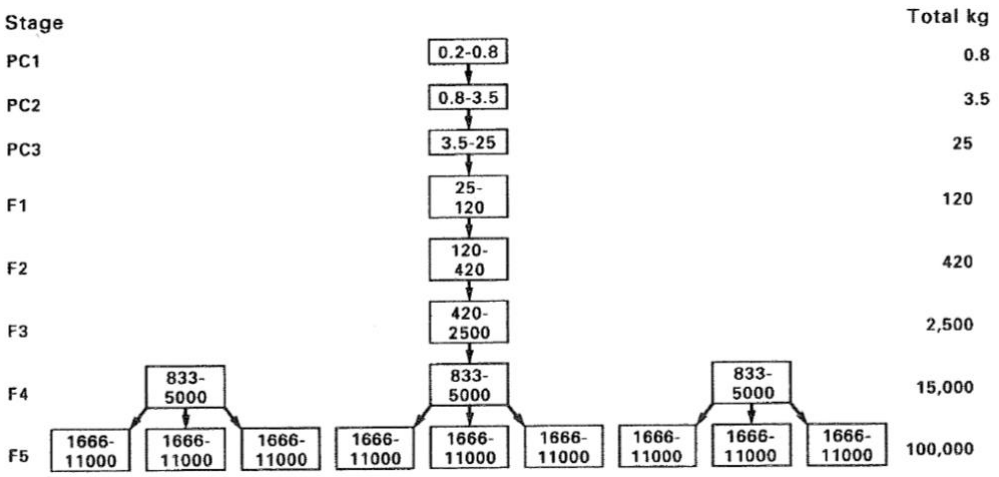
Current and past bt603 solved past papers:

Mcqs kch past me sy b thy....subjective past me sy aya..bx 2 3 questions different thy..

Flow chart of inocula. 10 marks:

Figure showing the development of the



Bull et al 5 pattern of form forming:

Hall *et al.* (1973) have recognized five patterns of foaming in fermentations:

1. Foaming remains at a constant level throughout the fermentation. Initially it is due to the medium and later due to microbial activity.

2. A steady fall in foaming during the early part of the fermentation, after which it remains constant. Initially it is due to the medium but there are no later effects caused by the micro-organism.

3. The foaming falls slightly in the early stages of the fermentation then rises. There are very slight effects caused by the medium but the major effects are due to microbial activity.

4. The fermentation has a low initial foaming capacity which rises. These effects are due solely to microbial activity.

...flow chart waly 2 question thy..

.physical characteristics in identification of microorganisms.

. Trace elements... 2

names of sample collection organization.

National Collection of
Type cultures (NCTC)
PHLS Central Public Health Laboratory, 61
Colindale Avenue, London NW9 5HT, UK
National Collections of
Industrial and Marine
Bacteria Ltd. (NCIB,
NCMB)
23 St Machar Drive, Aberdeen AB2 1RY, UK
National Collection of
Yeast Cultures (NCYC)
AFRC Institute of Food Research, Norwich
Laboratory, Colney Lane, Norwich NR4

7UA, UK

American Type Culture
Collection (ATCC)
12301 Parklawn Drive, Rockville, MD
20852, USA

.methods of foreign DNA in organisms..

Write equation of oxidation of glucose

Aeration & Agitation

The rate of oxygen transfer from air bubl

$$dC_L/dt = K_L a(C^* - C_L)$$

ase may

where C_L is the concentration of dissolved oxygen in the fermentation broth (mmoles dm^{-3}),

t is time (hours),

dC_L/dt is the change in oxygen concentration over a time period, i.e. the oxygen-transfer rate (mmoles O_2 dm^{-3} h^{-1}),

K_L is the mass transfer coefficient (cm h^{-1}),

a is the gas/liquid interface area per liquid volume (cm^2 cm^{-3}),

C^* is the saturated dissolved oxygen concentration (mmoles dm^{-3}).

Genetic technique for characterization of microbe

Genotypic characteristics

- **Nucleic acid probes**
- **DNA amplification**
- **Ribotyping**

Genotypic Characterization (Nucleic acid probes)

•Used to locate unique sequences

➢**Single-stranded DNA (or RNA)**
•**Generally labeled (radioactive or fluorescent)**
➢**Complementary to the sequence of interest**
➢**Observe and identify intact microorganisms**
•**Fluorescence in situ hybridization (FISH)**
➢**Observe and identify samples**
•**DNA amplification by PCR**
➢**Specific primers are used**

Phenotypic technique for microbe characterization

Phenotypic characteristics
- **Microscopic morphology**
- **Metabolic differences (Biochemical Testing)**
- **Serology**
- **Fatty acid analysis**

Aik graph wala kuch tangent wala tha i think oxygen balance technique waly topic sy tha k graph mein tangent ka show kr rha hai

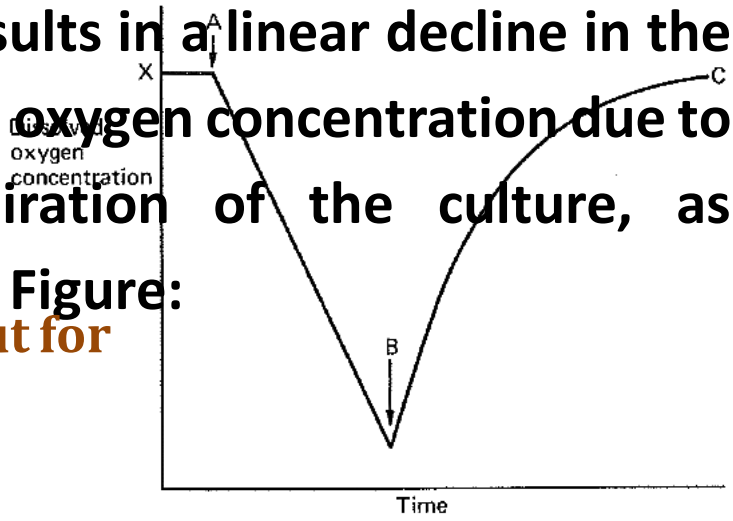
Aik dynamic gass out method wala graph aya tha

Dynamic Method of Gassing-out
Respiratory activity of a growing culture in the fermenter is utilized to lower the oxygen level prior to aeration. Therefore, the estimation has the advantage of being carried out during a fermentation which should give a more realistic assessment of the fermenter's efficiency.

Aeration & Agitation

The procedure involves stopping the supply of air to the fermentation which results in a linear decline in the dissolved oxygen concentration due to the respiration of the culture, as shown in Figure:

Dynamic gassing out for



Methods for foreign Dna in organism

Methods involving DNA foreign to the organism (i.e. recombination)

2. Transduction

3. Conjugation

4. Transformation

6. Protoplast fusion

7. Genetic engineering

8. Metabolic engineering

9. Site-directed mutagenesis

Streak plate method 10 marks

The isolation method most commonly used to get pure cultures is the streak plate method (Figure on next slide). A sterile inoculating loop is dipped into a mixed

culture that contains more than one type of microbe and is streaked in a pattern over the surface of the nutrient medium. As the pattern is traced, bacteria are rubbed off the loop onto the medium. The last cells to be rubbed off the loop are far enough apart to grow into isolated colonies. These colonies can be picked up with an inoculating loop and transferred to a test tube of nutrient medium to form a pure culture containing only one type of bacterium

Streak plate method for pureculturing of required microorganism is done for nonfilamentous unicellular microorganisms only.

The most desirable isolation medium would be one which selects for the desired types and also allows maximum genetic expression. Cultures grown on such media could then be used directly in a screen

However, it is more common that, once isolated, the organisms are grown on a range of media designed to

enhance
productivity.

Flow chart for using microorganism in industry General and specific inhibtor

Specific and General Inhibitors used in fermentations

Product	Inhibitor	Main effect	Micro-organism
Glycerol	Sodium bisulphate	Alcohol dehydro-duction repressed	*Saccharomyces cerivisiae*
Tetracycline	Bromide	Chlortetracycline formation repressed	*Streptomyces aureofaciens*
Glutamic acid	Penicillin	Dell wall permeability	*Micrococcus glutamicus*
Citric acid	Alkali metal/ phosphate, pH below 2.0	Oxalic acid repressed	*Aspergillus niger*
Valine	Various inhibitors	Various effects with different inhibitors	*Brevibacterium roseum*
Rifamycin B	Di-ethyl barbiturate	Other rifamycins inhibited	*Nocardia mediterranel*

Trace elements

The role of trace elements in medium formulation can be significant. Cultured cells normally require Fe, Zn, Cu, Se, Mn, Mo and V. These are often present as impurities in other media components. Cleveland *et al.* (1983) found that if the number of trace elements were increased, insulin, transferrin, albumin and liposomes were not needed in a serum-free hybridoma medium. They included Al, Ag, Ba, Br, Cd, Co, Cr, F, Ge, J, Rb, Zr, Si, Ni and Sn as well as those previously mentioned.

BT 603 – current solved past paper

- **Fermentation actually comes from Latin word “fervere” which means “to boil”**

- Fermentation can be classified into different types on the basis of followings:
    - *Oxygen (O_2) Need*
    - *Production Scale*
    - *State/Mixing*
    - *All of above*
- **Pilot and industrial scale fermentations** are normally carried out in stirred, aerated vessels
- **Examples of Bioweapons**
    - Smallpox
    - Anthrax
    - Botulinum toxin
    - All of above
- **Constitutive enzyme:** An enzyme that is synthesized continually regardless of growth conditions.
- In 1900-1940 Citric acid is produced
- **On the basis of state or mixing fermentation can be Shake-Flask Fermentation**
- **Downstream Processing** includes the **recovery of the products** in a pure state and the **effluent treatment**.
- **A medium with a high viscosity will also need a higher power input for effective stirring**
- **Reducing media contain ingredients, such as sodium thioglycolate, that chemically combine with dissolved oxygen and deplete the oxygen in the culture medium.**
- **Cellular yield coefficient of n-alkanes 1.03**
- ***Carbohydrate composition of barley malt (expressed as % dry weight of total) of Reducing sugars 3-4***
- ***Trace element Mn influencing primary and secondary metabolites production of Bacitracin***

- **In processes used for the production of glutamic acid, limited concentrations of biotin must be present in the medium**
- **Phenylacetic acid is still the most widely used precursor in penicillin production.**
- **potential chlorination inhibitors, e.g. bromide,**
- **heterologous proteins may show some degree of toxicity to the host and have a major influence on the stability of heterologous protein expression.**
- **The individual components of the medium can influence the viscosity of the final medium and its subsequent behaviour with respect to aeration and agitation.**
- **Trace elements were increased; insulin, transferrin, albumin and liposomes were not needed included Co**
- **The foaming can cause removal of cells from the medium which will lead to autolysis**
- **The foam is unavoidable and antifoam should be used.**
- **many antifoams are of low solubility**
- **independent variable (nutrient, antifoam, pH, temperature, etc.)**
- **When more than five independent variables are to be investigated, the Plackett-Burman design may be used**
- **The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth.**
- **The solubility of oxygen in pure water is 8 mg/L at 4°C.**
- **Production of 100 grams of biomass from hydrocarbon requires approximately three times the amount of oxygen to produce the same amount of biomass from carbohydrate.**
- **maximum biomass production may be achieved by satisfying the organism's maximum specific oxygen demand by maintaining the dissolved oxygen concentration greater than the critical level**
- **Sulphite Oxidation technique does not require the measurement of dissolved oxygen concentrations**
- **Two methods have been employed to achieve this lowering of the dissolved oxygen concentration - the static method and the dynamic method.**

- **(NCTC) abbreviation of? National Collection of Type cultures**
- **The isolation method most commonly used to get pure cultures is the streak plate method**
- **lyophilization (freeze-drying)**
- **the process of**
- **strain improvement involves the continual genetic modification of the culture,**
- **Protoplast fusion has been demonstrated as an efficient way to induce hetero-karyon formation and recombination with high frequency.**
- **Auxotrophy is the inability of an organism to synthesize a particular organic compound required for its growth**
- **The quantity of inoculum normally used is between 3 and 10% of the medium volume**

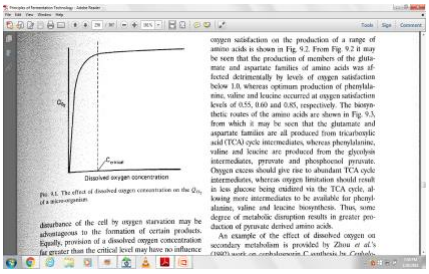
Q:- what is Site-Directed Mutagenesis? (2)

- Site-specific mutation commonly involve using mutagenic oligonucleotides in a primer extension reaction with DNA polymerase.
- This method allows for point mutation, or deletion or insertion of small stretches of DNA to be introduced at specific sites.

Q: Two examples of Methylotrophic yeasts?

Methylotrophic yeasts such as *Hansenula polymorpha* and *Pichia pastoris*

Q: Draw a graph showing The effect of dissolved oxygen concentration on the Q_{O2} of a microorganism?



oxygen satisfaction on the production of a range of amino acids is shown in Fig. 9.2. From Fig. 9.2 it may be seen that the production of members of the glutamate and aspartate families of amino acids was affected detrimentally by levels of oxygen satisfaction below 1.0, whereas optimum production of phenylalanine, valine and leucine occurred at oxygen satisfaction levels of 0.55, 0.60 and 0.85, respectively. The biosynthetic routes of the amino acids are shown in Fig. 9.3, from which it may be seen that the glutamate and aspartate families are all produced from tricarboxylic acid (TCA) cycle intermediates, whereas phenylalanine, valine and leucine are produced from the glycolysis intermediates, pyruvate and phosphoenol pyruvate. Oxygen excess should give rise to abundant TCA cycle intermediates, whereas oxygen limitation should result in less glucose being oxidized via the TCA cycle, allowing more intermediates to be available for phenylalanine, valine and leucine biosynthesis. Thus, some degree of metabolic disruption results in greater production of pyruvate derived amino acids.

An example of the effect of dissolved oxygen on secondary metabolism is provided by Zhou *et al.*'s

disturbance of the cell by oxygen starvation may be advantageous to the formation of certain products. Equally, provision of a dissolved oxygen concentration far greater than the critical level may have no influence

Q: Define Henry's Law?

$$C^* = \frac{P_o}{H}$$ ***Henry's Law describes the solubility of O_2 in nutrient solution in relation to the O_2 partial pressure in the gas phase***

C* is the oxygen saturation concentration of the nutrient solution,

P_o is the partial pressure of the gas in the gas phase

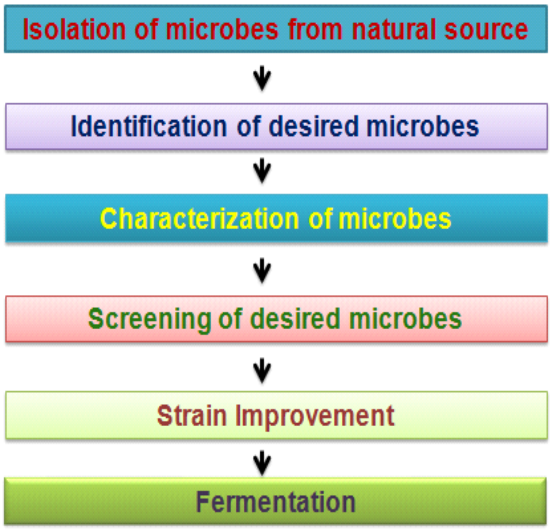
H is Henry's constant, which is specific for the gas and the liquid phase.

Q: Bull *et al.* (1979) cited criteria in the choice of process organism? (10)

Q: What is Replica Plating Method? (5)

Mutants can be detected by selecting or testing for an altered phenotype. Positive selection involves the selection of mutant cells and rejection of non-mutated cells. To identify mutants that have gained some physical trait, Replica plating is used for negative selection. Negative selection identifies mutants that have lost some physical trait - for example, auxotrophs that have nutritional requirements not possessed by the parent (nonmutated) cell.

Q: Draw a flow chart for INDUSTRIAL STRATEGY FOR USAGE OF MICROORGANISMS? (5)



Q: Phenotypic characteristics to characterize and identify microorganisms (3)?

- **Microscopic morphology**
- **Metabolic differences (Biochemical Testing)**
- **Serology**
- **Fatty acid analysis**

Q: Name Trace elements use in fermentation media, any five (3)

The role of trace elements in medium formulation can be significant. Fe, Zn, Cu, Se, Mn, Mo and V. Al, Ag, Ba, Br, Cd, Co, Cr, F, Ge, J, Rb, Zr, Si, Ni and Sn

a
2. Types,only name of water need fermentation
n the basis of
water/ moisture
requirement ,
process can be:

- **Solid State**
- **Submerged/ Liquid**
- **Surface Culture/ Jelly**

3. Formula for calculating corbon source.Midterm solved ppr
4. Ph long question

5. Difrence in situ ex situ
6. Name oxygen need base fermentation.
7. streak plate methode 10 marks

The isolation method most commonly used to get pure cultures is the streak plate method (Figure on next slide). A sterile inoculating loop is dipped into a mixed culture that contains more than one type of microbe and is streaked in a pattern over the surface of the nutrient medium. As the pattern is traced, bacteria are rubbed off the loop onto the medium. The last cells to be rubbed off the loop are far enough apart to grow into isolated colonies. These colonies can be picked up with an inoculating loop and transferred to a test tube of nutrient medium to form a pure culture containing only

one type of bacterium.
Streak plate
method for pureculturing of
required
microorganism is
done for nonfilamentous
unicellular
microorganisms
only.

8. air bubble
9. write and explain equation forign dna in organism
10. Ams test..? 2**Ames Test**
The Ames test is a relatively inexpensive and rapid test for identifying possible chemical carcinogens.
The test assumes that a mutant cell can revert to a normal cell in the presence of a mutagen and that many mutagens are carcinogens.
Histidine auxotrophs (require histidine in the media) of *Salmonella* are exposed to an enzymatically treated potential carcinogen, and reversions to the nonmutant state are selected.
11. Lyophilization..? 3**During lyophilization (freeze-drying), a suspension of microbes is quickly frozen at temperatures ranging from -54°C to -72°C, and the water is removed by a high vacuum (sublimation). While under vacuum, the container is sealed by melting the glass with a hightemperature torch. The remaining powder like residue that contains the surviving microbes can be stored for years. The organisms can be revived at any time by hydration with a suitable liquid nutrient medium.**
12. Equation of oxygen transfer from air bubble to liquid medium..? 5

The rate of oxygen transfer from air bubl

$$dC_L/dt = K_L a(C^* - C_L)$$

ase may

where C_L is the concentration of dissolved oxygen in the fermentation broth (mmoles dm^{-3}),

t is time (hours).

dC_L/dt is the change in oxygen concentration over a time period, i.e. the oxygen-transfer rate (mmoles O_2 dm^{-3} h^{-1}),

K_L is the mass transfer coefficient (cm h^{-1}),

a is the gas/liquid interface area per liquid volume (cm^2 cm^{-3}),

C^* is the saturated dissolved oxygen concentration (mmoles dm^{-3}).

13. Ek question bull 1992 contributions that(3)

Bull (1992) summarized the major contributions as follows:

1. The provision of test organisms that have increased sensitivities, or resistances, to known agents. For example, the use of super-sensitive strains for the detection of β-lactam antibiotics.

2. The cloning of genes coding for enzymes or receptors that may be used in inhibitor or binding screens makes such materials more accessible and available in much larger amounts.

3. The development of reporter gene assays. A reporter gene is one which codes for an easily assayable product so that it can be used to detect the activation of a control sequence to which it is fused.

14. What is malt.? 2mrks

Barley grains may be partially germinated and heat treated to give the material known as malt, which contains a variety of sugars besides starch. Malt is the main substrate for brewing beer. Malt extracts may also be prepared from malted grain.

15. Exsitu and insitu sterilization.? 2mrks

16. Name Types of fermentation on the basis of oxygen need..? 3

The O_2 supply must be according to need of the process of fermentation:

- **Aerobic Fermentation**
- **Semi-anaerobic Fermentation**
- **Anaerobic Fermentation**

❑Aerobic Fermentation

➢Aerobic fermentation is carried out in the presence of oxygen.

➢Organisms use oxygen for the conversion of complex organic compounds.

❑Semi-anaerobic Fermentation

Process in which organism makes ATP by aerobic respiration , but capable of switching to anaerobic respiration if oxygen is absent.

❑Anaerobic Fermentation

Anaerobic fermentation is carried out in the **strict absence of oxygen**; means presence of oxygen may be toxic.

17. Limitations of starch..? 3

18. Types of fermenation on basis of water content/moisture and define each type..? 5

19. Explain 5 ways to control pH in fermentation medium..? 5

20. Phenotypic technique?

21. Genotypic technique?
22. Describe henry law?
23. What is bull criteria?
24. Name of specific inhibitors?
25. Why medium is on inhibitors, give two reason esa he kch tha
26. Name of specific inhibitors
27. Method of screening antibiotics?
28. Why antifoam is excess in media give three methods?

Difrence b/w differential and reducing media Difrence b/w differential and reducing media